

Neural Network Controller for Boost Converter

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Abstract—This paper presents a neural network controller scheme for boost converters. This controller is designed to stabilize the output voltage of the boost converter and to improve its performance during transient operations. To investigate the effectiveness of the proposed controller, certain operations such as starting up and reference voltage variations are verified. The simulation results show that the proposed controller has an improved performance compared to the conventional PI controller method.

Key Words: - Boost converter, Neural network, Performance comparison, MATLAB

I. NOMENCLATURE

$d(N)$	- Duty cycle
$\delta d(N)$	- Change of duty cycle
V_i	- Input voltage source
V_o	- Output voltage(V)
SW	- Switching component used in equivalent circuit of boost converter
D	- Diode used in equivalent circuit of boost converter
C	- Capacitance used in equivalent circuit of boost converter
L	- Inductance used in equivalent circuit of boost converter
R	- Load resistance
f	- PWM switching frequency
u	- Binary control switch
R	- Output voltage ripple
V_{ref}	- Reference voltage for the ANN controller.

II. INTRODUCTION

DC power supplies have extensive applications in wide areas ranging from simple electronic devices such as laptops to much more advance electric vehicles and aerospace industries. The DC-DC converter is used to convert a given dc voltage level to a different dc voltage level in order to satisfy the voltage requirements of the load. The DC-DC converters are extensively employed in power conditioning of the non-conventional electrical energy sources such as photovoltaic, wind generators and fuel cell systems. Due to these reasons, the DC-DC converters have a more potential market in the future [1]-[2].

Basically, the DC-DC converters are made up of power semiconductor devices which are operated as electronic

switches and is classified as switched-mode DC-DC converters or normally referred to as Switched mode power supply (SMPS). Nowadays to improve the efficiency and performance different switching circuit topologies are used. In order to maintain a stable output in a DC-DC converter, an appropriate control signal should be applied. Conventional controller designs assumed all circuit components as ideal with no power loss and performance degradation and the circuit is operated at a stable bias point so that it can be modeled by a set of linear equations. However, in practice the switching network is highly nonlinear. An accurate mathematical model of the switching network is very difficult to obtain. In addition there are also reported problems of the supply voltage and load current fluctuating over a wide range. A real-time adaptive control thus becomes necessary to achieve a proper system performance [3]-[4].

The switching operation causes the occurrence of inherent nonlinear characteristics of converters. Due to undesired nonlinear characteristics, the DC-DC converters require a controller with a high degree of dynamic response. Pulse width modulation (PWM) technique is the often used switching control method. Since the past decade Proportional-Integral-Differential (PID) controllers are employed for PWM switching control mainly because of its simplicity. However, the implementation of this control technique to the nonlinear plants such as the power converters will undergo dynamic response in the converter output voltage regulation. In general, PID controller creates a longer rise time when the overshoot in the output voltage decreases. To overcome this and improve the dynamic response of the DC-DC converters, a number of intelligence controllers such as fuzzy logic control, neural network control and hybrid neuro-fuzzy control methods are adopted [4].

In recent times artificial neural networks (ANN) have been applied to improve the performance of DC-DC converters in dynamic systems to get performances with good regulation. ANN is a type of intelligence control which has the ability to approach any function by learning process, hence this controller is suitable for nonlinear control system. ANN based control provides an excellent dynamic characteristics of DC-DC converter when compared with the conventional control technique [5]-[7].

III. DC-DC CONVERTERS AND PULSE WIDTH MODULATION TECHNIQUES

The DC-DC converters can convert a certain DC voltage level to another by switching action. The basic DC-DC converter topologies are shown in Fig.1.

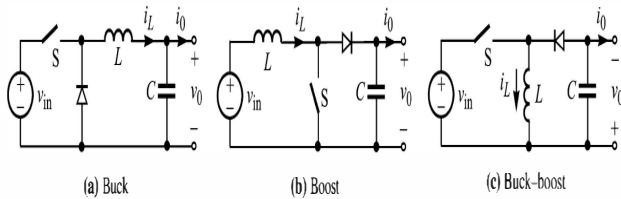


Fig.1. Basic DC-DC Converter topologies

The output voltage is controlled by adjusting the ON time of the switch, which in turn adjusts the pulse width of the output voltage. This is known as pulse width modulation (PWM). By varying the duty ratio of the PWM modulator, the DC-DC converter can convert one level of electrical voltage to any desired level. Among various modulation techniques, constant-frequency trailing-edge PWM modulation with triangular carrier is the common technique employed in DC-DC converters. The different PWM methods are shown in Fig.2 and Fig.3.

For DC-DC converters, the PWM reference is constant when the converter operates in a steady state but varies as the converter goes through a transient state. The spectral characteristics can be readily determined by Fourier analysis. To characterize the dynamic behavior of a modulator in converters, the reference can be assumed to consist of a dc value corresponding to the steady-state duty ratio and a sinusoidal component representing a small-signal perturbation. A small-signal model can be obtained by computing the component in the PWM output at the frequency of the sinusoidal perturbation [8].

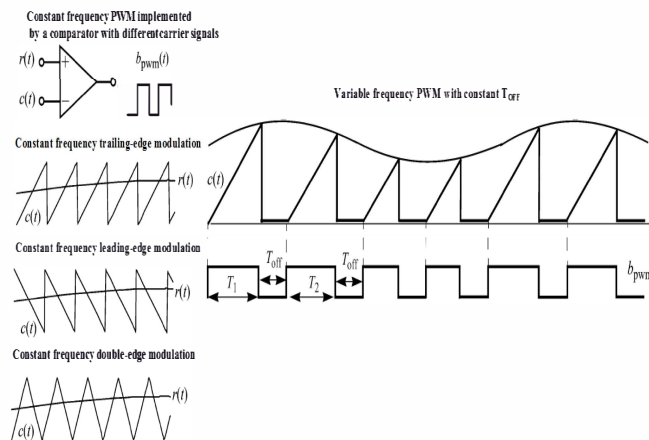


Fig.2. Basic PWM techniques-1

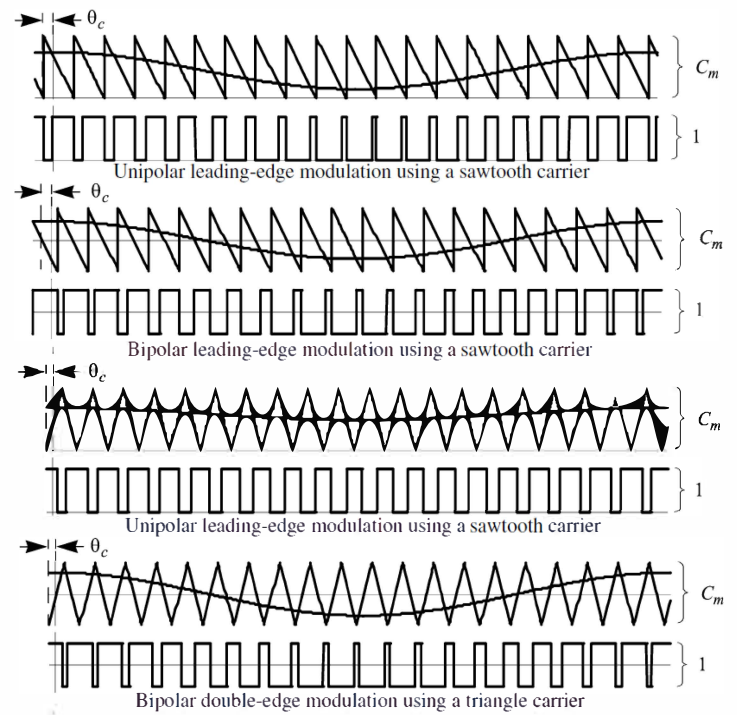


Fig.3. Basic PWM techniques-2

IV. CONTROLLERS FOR DC-DC CONVERTERS

Past decades, the controller design for DC-DC converters has been carried out through analog circuits, which limited them to use PI control structures. The selection of the controller parameters in the PI controller is a compromise between robustness and transient response. There have been many approaches proposed for PWM switching control design based on the sliding-mode control, fuzzy control and adaptive neural control techniques. In the sliding-mode control design for DC-DC converters, the controllers are simple to design and easy to implement. However, their performances normally depend on the operating point, thus these control parameters which are required to ensure proper behavior in all operating conditions are hard to design. In the fuzzy control design for DC-DC converters, the fuzzy rule should be pre-selected through trial-and-error to achieve satisfactory performances. This is time consuming. Even though the usage of online tuning algorithm provided satisfactory regulator performance, the computation loading of these learning algorithms is too complex. The neural-network-based control technique has been represented as an unconventional design method for various control systems. The successful key point is its approximation ability in which it can approximate an unknown system dynamics or an ideal controller after learning. Based on this property, ANN controllers have been developed to compensate for the effects of nonlinearities and system uncertainties, so that the stability, convergence and robustness of the control system can be improved [9]-[11].

V. DEFINITION OF THE PROBLEM

DC-DC converter is very popular because of its high efficiency and compact size. PWM is the most widely used

duty ratio control method where switching frequency is constant and the duty cycle, $d(N)$, varies with the load resistance variations at the N^{th} sampling time. The output of the designed controller, $\delta d(N)$, is the change in duty cycle. Then, the duty cycle is determined by adding the previous duty cycle $d(N-1)$ to the change in duty cycle $\delta d(N)$, i.e. $d(N) = d(N-1) + \delta d(N)$. This duty cycle signal is then sent to a PWM output stage that generates the appropriate switching pattern for the switching power supplies.

Artificial Neural Network (ANN)

ANNs are extensively established as a technology offering, an alternative way to solve complex problems because; these ANNs do not require any detailed information about the system. They learn the relationship between the input and output variables by studying the previously recorded data. Another benefit of using ANNs is their capability to handle large and complex systems with many consistent parameters. These trained ANNs can be used to approximate an arbitrary input-output mapping of the system. There are many artificial neural network architectures. One such which is predominant is the feedforward neural network (FNN). The standard neuron structure illustrated in Fig. 4 is taken up which includes a summer and a logistic function $f(\text{net}_{p,i})$ that can be either a sigmoid or a linear function [5]- [7].

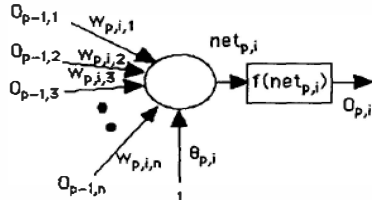


Fig. 4. Structure of an elementary neuron

The equation for the i^{th} neuron of the p^{th} layer structure is

$$\text{net}_{p,i} = \sum_{k=1}^n w_{p,i,k} O_{p-1,k} + \theta_{p,i} \quad (1)$$

$$O_{p,i,k} = f(\text{net}_{p,i}) \quad (2)$$

Where, $O_{p,i,k}$ is the output, $O_{p-1,k}$ is the k^{th} output at the $(p-1)^{\text{th}}$ layer, $w_{p,i,k}$ is the weight from the k^{th} input of the $(p-1)^{\text{th}}$ layer to the i^{th} output of the p^{th} layer, and $\theta_{p,i}$ is the bias. These neurons are organized in layers as shown in Fig. 5.

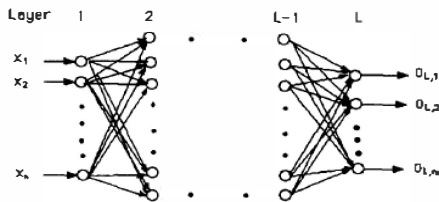


Fig. 5. Feed forward neural network architecture

Scaled data enter the network at neurons of the input layer and are propagated to the output through intermediate layers. Each connection has associated with it a weight which acts to modify the signal strength. There are different methods for optimizing criterion functions to train the neural network. The training of neural network usually requires iterative techniques. The back-propagation method is the trendiest

technique trained by using the gradient descent method among the available training algorithms because of its simplicity, stability, robustness, and ease of implementation, but it has a disadvantage of slow convergence speed.

ANN control based boost Converter

In this paper, boost converter with ANN based control is presented. Feed forward neural network with back propagation training has been selected. A gradient descent minimization can be performed on the error function. Fig. 6 shows the basic circuit diagram of boost converter.

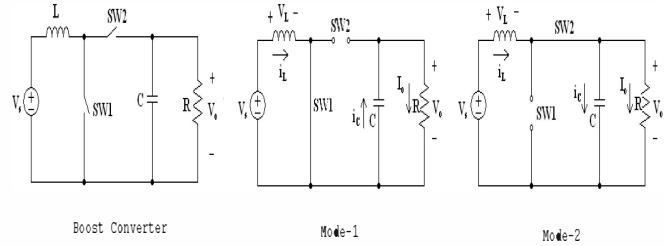


Fig.6. Boost converter with its equivalent circuits

Boost converter is a step up converter. Regulation is achieved by varying the duty cycle of the PWM input at fixed frequency. When the duty cycle (d) varies from 0 to 1, the output voltage of converter is varied. Let us assume that converter is operated in continuous current mode (CCM).

During mode-1 (when SW1 is ON), $V_L = V_s$. During mode-2 (when SW1 is OFF), $V_L = V_s - V_o$. The expression for duty cycle is, $D = 1 - (V_s/V_o)$ for CCM and transfer function can be written as $V_o/V_s = 1/(1-D)$. The expression for CCM inductance is

$$L_{\text{CCM}} = \frac{D(1-D)^2 R}{2f} \quad (3)$$

The largest inductance occurs with maximum resistance. The duty ratio which will give the maximum inductance value is obtained by differentiating the above equation and equating it to zero. The values are 1/3 and 1. $D=1$ is not practical, so L_{max} can be obtained when $D=1/3$. A small change in D results in large variation in L value. A proper value of L should be designed for CCM operation. Same way CCM capacitance is calculated from

$$C_{\text{CCM}} = \frac{D}{Rf} \quad (4)$$

The conventional model of converters depicts the behavior of converter without dynamic characteristics and presents no indication of how the converter behaves under transient conditions. Once the inductance and capacitance values are determined using equations mentioned above, they are used in simulation to observe the transient behavior of the circuit to understand the effects of increase or decrease in L and C .

The mathematical transient features of a circuit must be expressed using differential equations. To facilitate the simulation process, the differential equations are formulated into the state space model of the circuit. The state variables

are voltage and current in converter circuits. The number of inductors and capacitors determine the number of state variables and hence the order of the state-space system. The sources in the circuit are the forcing functions and comprise of the control vector. In general, the number of sources m , determines the dimensions of the control vector and the control input matrix.

The circuit topologies of the boost converter [4], shown in Fig.6 are redrawn in Fig.7 as,

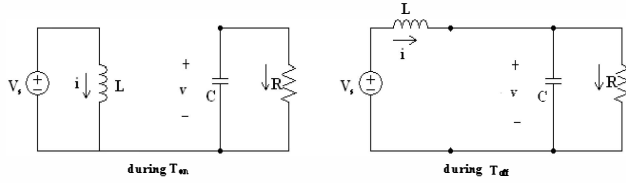


Fig.7. Boost converter with its equivalent circuits

With reference to Fig.7, during ON period, i.e., $0 \leq t \leq t_{on}$,

$$V_s = L \frac{di}{dt} \text{ and } 0 = C \frac{dv}{dt} + \frac{v}{R} \quad (5)$$

These equations describe the dynamics of the boost converter during ON period.

With reference to Fig.7, during OFF period, i.e., $t_{on} \leq t \leq T$,

$$V_s = L \frac{di}{dt} + v \text{ and } i = C \frac{dv}{dt} + \frac{v}{R} \quad (6)$$

(5) and (6) are combined by the definition of u as follows:

$$u = \begin{cases} 1, 0 \leq t \leq t_{on} \\ 0, t_{on} \leq t \leq T \end{cases} \quad (7)$$

Hence the equations are rewritten as,

$$V_s = L \frac{di}{dt} + (1-u)v \text{ and } (1-u)i = C \frac{dv}{dt} + \frac{v}{R} \quad (8)$$

The solutions of (8) for the derivatives yield the dynamic equations of the boost converter throughout the switching period as,

$$\frac{di}{dt} = -\frac{1}{L}v + \frac{1}{L}V_s + \frac{1}{L}vu \quad (9)$$

$$\frac{dv}{dt} = \frac{1}{C}i - \frac{1}{RC}v - \frac{1}{C}iu \quad (10)$$

With the source voltage and binary switch treated as separate control inputs, the variable structure state-space model of the boost converter is formulated as

$$\frac{d}{dt} \begin{bmatrix} i \\ v \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i \\ v \end{bmatrix} + \begin{bmatrix} \frac{V_s}{L} \\ -\frac{i}{C} \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \begin{bmatrix} u \\ V_s \end{bmatrix} \quad (11)$$

The state space model of the boost converter has two forcing functions and the control matrix does not have constant elements, it is the function of state variables.

Simulation of state space model in MATLAB

MATLAB is an excellent platform for state space model simulation because of the matrix data structure on which it is based. MATLAB also has function files that compute numerical solutions to coupled differential equations. The

m-file [4] contains the control function and state derivative function described by

$$\dot{\bar{x}} = A\bar{x} + B\bar{u} \quad (12)$$

The *ode23t* solver has been chosen which uses variable step length to improve computational accuracy. Unfortunately, the values of physical components typically result in large magnitude derivatives and longer simulation times. However this issue is readily addressed if the differential equations and state space models are normalized with respect to switching frequency [4]. The normalization process alleviates the requirement to compute large derivatives and significantly decreases the simulation time. Normalization with respect to frequency: Since the unit of time is second and the unit of frequency is the inverse of a second, a normalization results from the product of frequency and time. That is, $\tau = ft$ and the solution for time is,

$$t = \frac{\tau}{f} \quad (13)$$

Differentiation (13) results in

$$\frac{dt}{d\tau} = \frac{1}{f} \quad (14)$$

A differential equation is normalized upon substitution for t and dt using the above equations.

The normalized converter model simulates significantly faster than the time based model. The normalized state space model of the boost converter from (11) is given as

$$\frac{d}{d\tau} \begin{bmatrix} i \\ v \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{Lf} \\ \frac{1}{Cf} & -\frac{1}{RCf} \end{bmatrix} \begin{bmatrix} i \\ v \end{bmatrix} + \begin{bmatrix} \frac{V_s}{Lf} \\ -\frac{i}{C} \end{bmatrix} + \begin{bmatrix} \frac{1}{Lf} \\ 0 \end{bmatrix} \begin{bmatrix} u \\ V_s \end{bmatrix} \quad (15)$$

The frequency used in the model is the PWM switching frequency. The simulation time is the number of integral periods. In normalized model, the division of model by f results in smaller magnitude derivatives and shorter simulation times. Bipolar double edge modulation with triangular carrier is used for PWM pulse generation. M-file code for boost converter with state space modeling is given in [4]. The simulation results are as shown in Fig.8. This result is for particular line voltage and load.

The data required for training the ANN is obtained by running the above program for different line and load changes. ANN is trained such that any change in the source or load side does not affect the regulation. For training, all data are normalized and the controlled output from ANN is de-normalized and given to PWM modulator to change the pulses to switch. The basic block diagram of the ANN control based boost converter is shown in Fig.9.

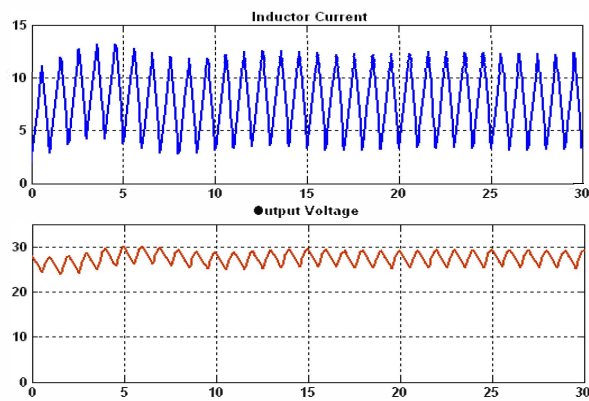


Fig.8. Simulated results for CCM Boost converter

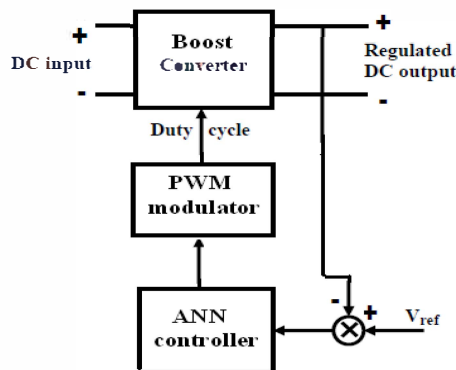


Fig.9. Block diagram of ANN controller based Boost regulator

Parameters of Neural Network:

No. of input data: 4 (Error, output voltage, input voltage and reference voltage)

Number of input neurons: 4

Number of hidden layers: 9

Number of output neuron: 1 (processed control output)

Number of data for training: 100

Stopping criterion: Error goal 0.002 or Maximum Epoch

Fig.10 shows the detailed schematic of ANN controller based boost converter.

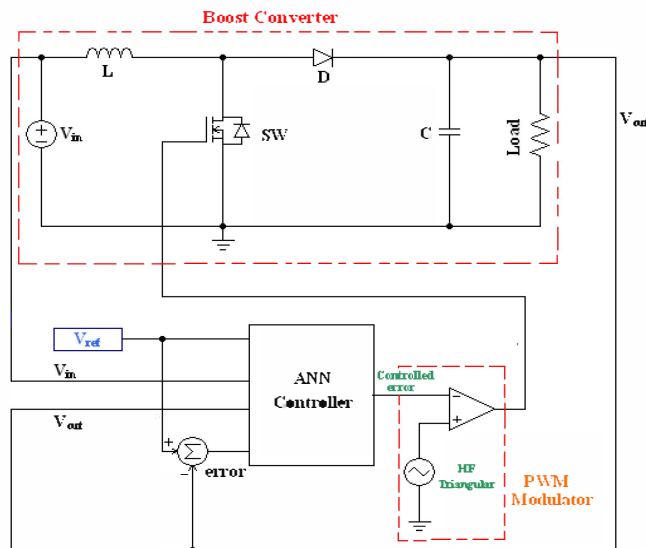


Fig.10. Schematic of ANN controller based Boost converter

Fig.11 shows the comparative results of the conventional PI controller and ANN controller performances.

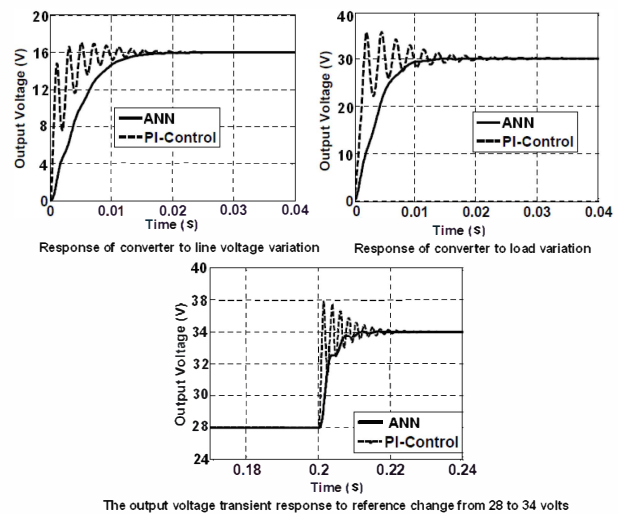


Fig.11. Comparison between PI controller and ANN controller

It is observed that the output voltage startup transient response of the boost converter with reference voltage is higher than that of the input voltage source. It is evident from Fig.11 that under the reference voltage-step variations, the ANN controller has a better transient response than those obtained from the PI controller.

VI. CONCLUSION

A neural network controller for DC-DC Boost converter has been presented in this paper. To improve the performance of the ANN controller, an online learning algorithm based on back propagation scheme is employed. The Simulation results shows that implementation of the online learning technique is feasible for the boost converter. The ANN controller here mimics the operation of PI controller. The ANN based controller proves to have a fast response in tracking the desired output voltage and is also effective in decreasing overshoot, oscillations and settling time.

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